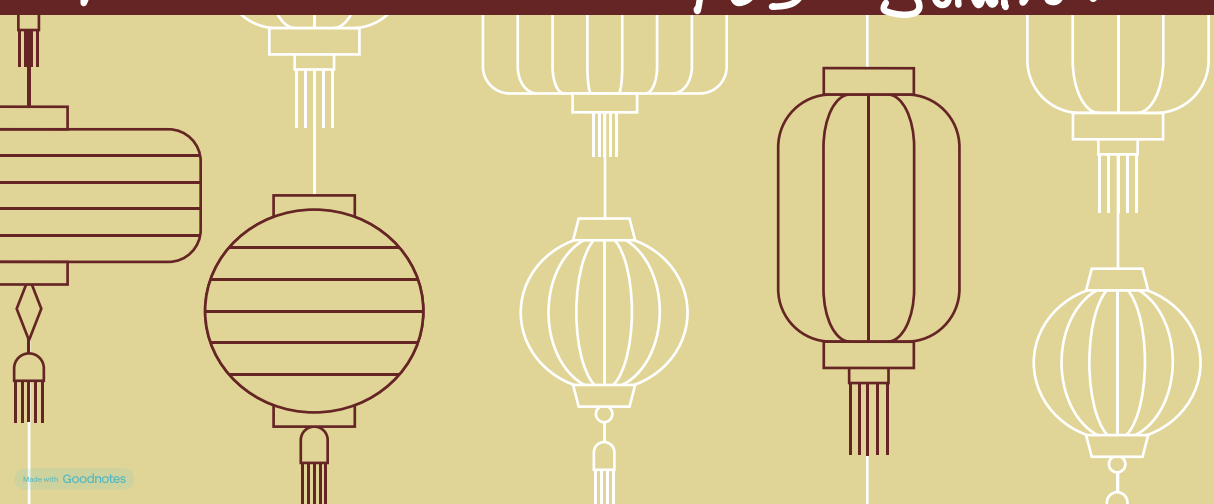
Cyrus Hong

Aurorcys

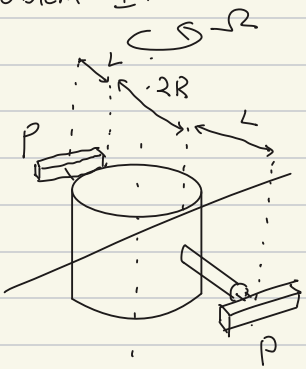
MIT 2.003 SC PS3 Solution



Made with Goodnotes

Problem 1:

a)



$\Omega = 0.05\ rad/s\ \hat{k}$

P instruments are symmetric

L initially $= 1.2m$

Once satellite deployed, L can change

L can be $0 \leftrightarrow 3m$

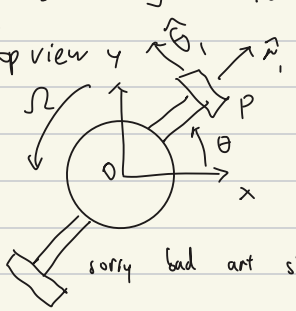
Maximum acceleration of P is $0.011\ m/s^2$

Determine maximum allowable boom extension rate

The acceleration of the satellite itself does not act any force on the instrument, as the centripetal force on the instrument (as the satellite goes around the earth) is exactly equivalent to the gravitational force of the earth on the instrument. No fictitious force would be felt by the instrument by the orbit of the satellite

Hence, finding $a_{P/O}$ is enough

Top view



sorry bad art skillz.

$$v_{c/o} = \left(\frac{\partial \rho_{c/o}}{\partial t}\right)_{\Omega=0} + \vec{\omega} \times \rho_{c/o}$$

$$a_{c/o} = a_{c/o}{}_{\Omega=0} + \vec{\alpha} \times \rho_{c/o} + \vec{\omega} \times (\vec{\omega} \times \rho_{c/o}) + 2\vec{\omega} \times v_{c/o}{}_{\Omega=0}$$

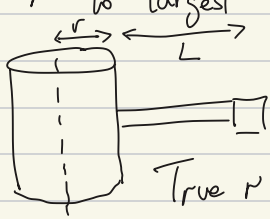
$$a_{c/o}{}_{\Omega=0} = \vec{\alpha} = 0$$

$$a_{c/o} = \Omega\hat{k} \times (\Omega\hat{k} \times r\hat{N}_1) + 2\Omega\hat{k} \times \dot{r}\hat{N}$$

$$= -\Omega^2 r\hat{N} + 2\dot{r}\Omega\hat{\theta}$$

$\therefore$ we want $|a_{P/O}| < 0.011\ \frac{m}{s^2}$

Set r to largest for worst case scenario



True $r = L + r$

$$\frac{d}{dt}(r\hat{N}) = \dot{r}\hat{N} + r\dot{\Omega}\hat{\theta}$$

$$\hat{r} = \cos\theta\hat{i} + \sin\theta\hat{j}$$

$$\frac{d}{dt}\hat{r} = \dot{\theta}(-\sin\theta\hat{i} + \cos\theta\hat{j})$$

$$= \dot{\theta}\hat{\theta} \quad \because \dot{\theta} = \Omega$$

$$\therefore = \Omega\hat{\theta}$$

Made with Goodnotes

$L_{longest} = 1.2m + 3m = 4.2m$

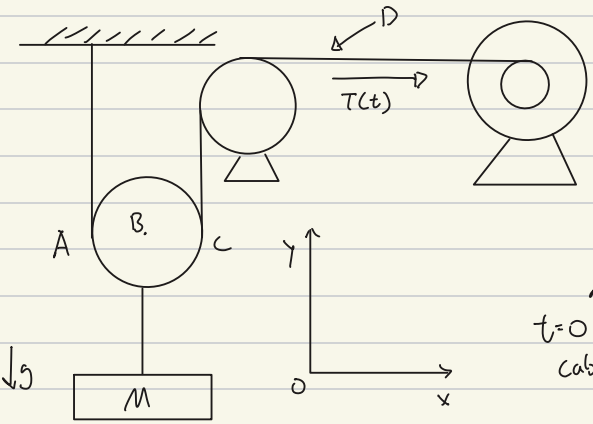
$\Omega = 0.05$ rad/s

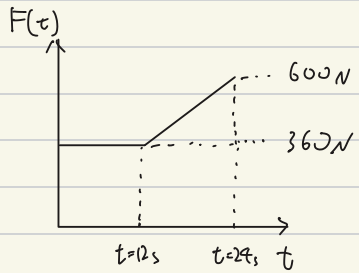
$$|a_{P/O}| = \left|-\Omega^2 r\hat{r} + 2\dot{r}\Omega\hat{\theta}\right|^2 \leq 0.01^2$$

$$= 0.05^4 \times 4.2^2 + 4\dot{r}^2 0.05^2 \leq 0.01^2$$

$$\dot{r} \leq 0.032787\ldots \frac{m}{s}$$

Problem 2



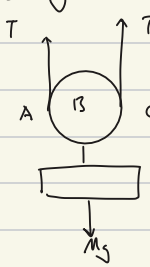


$M = 80\,kg$

$t = 0$, $v_{M/O} = 0$

cables & pulleys massless

Using $\sum F = ma$. We can find the integral of the acceleration



$$\sum_{ext} F = Ma = (2T - Mg)\hat{j}$$

$$a = \frac{2T}{M} - g$$

$$v = \int_0^t \left(\frac{2T}{M} - g\right) dt = \int_0^{12} \left(\frac{2T}{M} - g\right) dt + \int_{12}^{24} \left(\frac{2T}{M} - g\right) dt$$

$$= \left[\left(\frac{720}{M} - g\right)t\right]_0^{12} + \int_{12}^{24} \left(\frac{40t + 240}{M} - g\right) dt$$

$$= -9.72 + \int_{12}^{24} (0.5t - 6.81)\, dt$$

$$= -9.72 + \left[0.25t^2 - 6.81t\right]_{12}^{24}$$

$$= 16.56\ m/s$$

From $t = 12$ to $t = 24$, $F(t)$ graph:

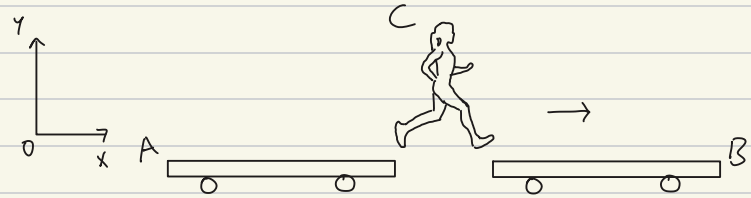
$$\frac{y - 360}{x - 12} = \frac{600 - 360}{24 - 12}$$

$$y = 20x + 120$$

Answer: $v_{M/O} = 16.56\ m/s\ \hat{j}$

Made with Goodnotes

Problem 3:



a: find $v_{A/0}$

b: $v_{B/0}$ after landing.

C girl is 75kg

$v_{C/A} = 3 m/s \, \hat{i}$

A, B both 50kg

A, B both originally at rest

$v_{C/0} + |v_{A/0}| = 3$

$\therefore v_{A/0} < 0$

$v_{C/0} - v_{A/0} = 3$, $v_{C/0} = 3 + v_{A/0}$

$0 = v_{C/0} \times 75 + v_{A/0} \times 50$

$0 = 225 + 75 \times v_{A/0} + 50 \times v_{A/0}$

$v_{A/0} = -1.8 ms^{-1} \, \hat{i}$

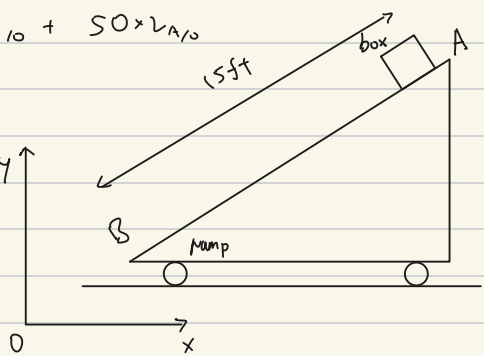
$v_{C/0} = 1.2 ms^{-1} \, \hat{i}$

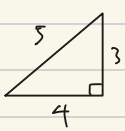
Using conservation of Linear Momentum

$v_{C/0} \times 75 = v_{B\&C/0} \times (50 + 75)$

$v_{B\&C/0} = 0.72 ms^{-1} \, \hat{i}$

Problem 4:





Pythagorean triple

Ramp: 120 lb ≈ 54.4 kg

box: 80 lb ≈ 36.3kg

15ft = 4.56m

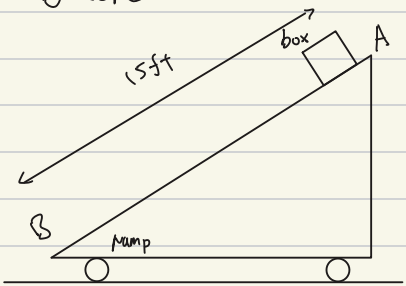
Property to use: COM remains at rest unless external force is acted upon it.

Step 1: draw before & after graph

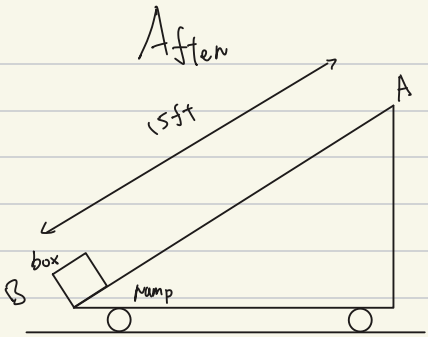
Step 2: solve with algebra

Made with Goodnotes

Before



After



Lets represent the x location of the COM of box & ramp in simple variables respectively

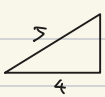
before: x_{box_1}, after: x_{box_2} (for box)

before: x_{ramp_1}, after: x_{ramp_2} (for ramp)

Initial COM x coord = Final COM x coord

$$(x_{box_1} m_{box} + x_{ramp_1} m_{ramp}) = (x_{box_2} m_{box} + x_{ramp_2} m_{ramp})$$

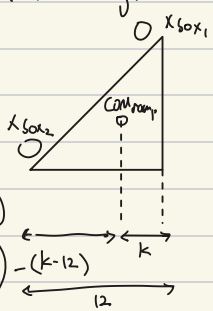
$$m_{box}(x_{box_1} - x_{box_2}) = m_{ramp}(x_{ramp_2} - x_{ramp_1})$$

∵ 

, 15 ft as hypotenuse, its bottom length is 12 ft

Assuming $x_{box_1} - x_{ramp_1} = k$ ... (1)

then $x_{box_2} - x_{ramp_2} = k - 12$ ... (2)



Now: (1) - (2):

$$(x_{box_1} - x_{box_2}) = 12 - (x_{ramp_2} - x_{ramp_1})$$

sub it in previous eq. (Let $(x_{ramp_2} - x_{ramp_1})$ be D)

$$m_{box}(12 - D) = m_{ramp} D$$

$$80(12 - D) = 120 D$$

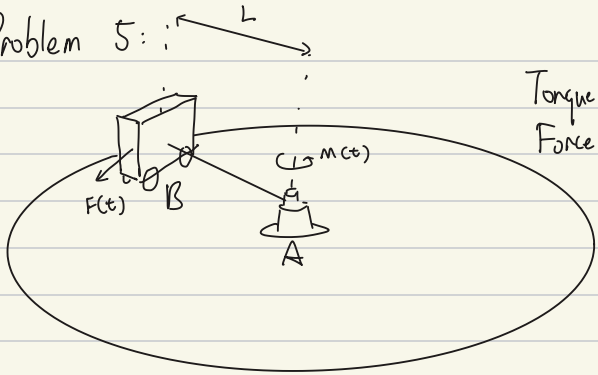
$$D = 4.8 \text{ ft}$$

$$(x_{ramp_2} - x_{ramp_1}) = 4.8 \text{ ft}$$

Ramp moves 4.8 ft right

Made with Goodnotes

Problem 5:



Torque: $M(t) = 30t^2$ Nm

Force: $F(t) = 15t$ N

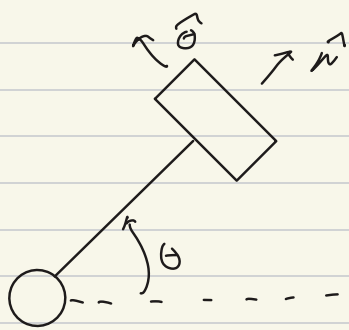
$t=0$ (start)

a) find angular momentum of B wrt A

b) $t=5$, $M(t) = F(t) = 0$, find $\vec{P}$, linear momentum of cart after switch off

c) Compute time derivative of $\vec{P}$ wrt $Oxyz$, explain the physical meaning of the force. $\frac{d}{dt}P = \Sigma F$.

a) Finding $h_{/A}$ (B just like a particle)



$$h_{/A} = r_{B/A} \times m v_{B/A}$$
$$= r_{B/A} \times m(\vec{\omega} \times r_{B/A})$$
$$= L\hat{r} \times m(\dot{\theta}\hat{k} \times L\hat{r})$$
$$= L\hat{r} \times m\dot{\theta}L\hat{\theta}$$
$$= mL^2\dot{\theta}\hat{k}$$

b) Using property: $\frac{d}{dt}(h_{/A}) = \sum_i \tau_{i/A}$

$$mL^2\dot{\theta}\hat{k} = \int_0^t (30t^2 + 15t \times L)\hat{k}\, dt$$

$\therefore t = 5$ at end

$$mL^2\dot{\theta}\hat{k} = \left[10t^3 + \frac{15 \times L}{2}t^2\right]_0^5 \hat{k}$$

$$mL^2\dot{\theta}\hat{k} = \left(1250 + \frac{375}{2}L\right)\hat{k}$$

$$\dot{\theta}\hat{k} = \left(\frac{1250}{mL^2} + \frac{375}{2mL}\right)\hat{k}$$

$$\dot{\theta} = \frac{1}{mL}\left(\frac{1250}{L} + \frac{375}{2}\right)\hat{k}$$

$$\dot{\theta} = \frac{1}{mL}\left(\frac{1250}{L} + \frac{375}{2}\right)\hat{k} \text{ rad/s}$$

$$\vec{P} = mv_{B/A} = m(\dot{\theta}\hat{k} \times L\hat{r}) = m(\dot{\theta}L\hat{\theta}) = \left(\frac{1250}{L} + \frac{375}{2}\right)\hat{\theta} \text{ kgms}^{-1}$$

Made with Goodnotes

c) $\frac{d}{dt}(\vec{p}) = m a_{B/A} = \frac{d}{dt}\left(m(L\dot{\theta}\hat{\theta})\right)$

$= \frac{d}{dt}(mL\dot{\theta}\hat{\theta})$

$= mL\ddot{\theta}\hat{\theta} - mL\dot{\theta}^2\hat{r}$

$\hat{\theta} = -\sin\theta\hat{i} + \cos\theta\hat{j}$

$\frac{d}{dt}\hat{\theta} = -\cos\theta\dot{\theta}\hat{i} - \sin\theta\dot{\theta}\hat{j}$

$= \dot{\theta}(-\hat{r})$

$= -\hat{r}\dot{\theta}$

$\ddot{\theta} = 0$

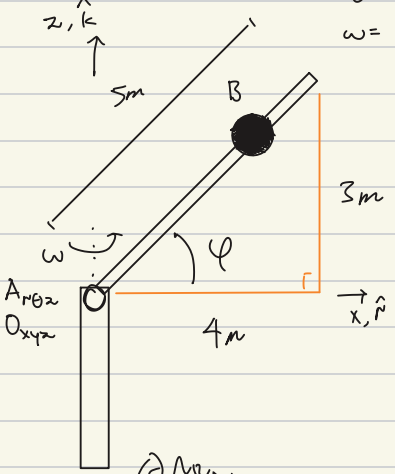
so $\frac{d}{dt}p = -mL\dot{\theta}^2\hat{r}$

This is the centripetal term to keep the block in a circular path.

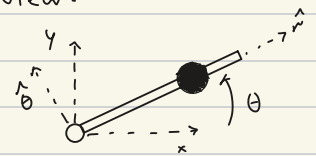
Problem 6:

B: monkey, 3kg, $v_{B/O}$ (with $\omega = 0$) $= 1.5\,ms^{-1}$

$\omega = 2.0\,rad/s$



Top view:



a) find $v_{B/O}$, $a_{B/O}$

b) find $h_{/O}$, the direction of $h_{/O}$ of monkey. Draw an arrow of it. Treating B as a particle

c) Find time derivative of $h_{/O}$ wrt $Oxyz$ frame.

a) $v_{B/O} = \left(\frac{\partial r_{B/O}}{\partial t}\right)_{\omega=0} + \omega_{/O} \times r_{B/O}$

$= (\dot{r}\hat{r} + \dot{z}\hat{k}) + \Omega\hat{k} \times (r\hat{r} + z\hat{k})$

$= \dot{r}\hat{r} + \dot{z}\hat{k} + \Omega r\hat{\theta}$

$\hat{\theta} = -\sin\theta\hat{i} + \cos\theta\hat{j}$

$\frac{d}{dt}\hat{\theta} = -\hat{r}\dot{\theta}$

$\hat{r} = \cos\theta\hat{i} + \sin\theta\hat{j}$

$\therefore \hat{r} = \cos\theta\hat{i} + \sin\theta\hat{j}$ & $\hat{\theta} = -\sin\theta\hat{i} + \cos\theta\hat{j}$ $\quad \frac{d}{dt}\hat{r} = \dot{\theta}\hat{\theta}$

$\therefore v_{B/O} = (\dot{r}\cos\theta - \dot{z}\sin\theta)\hat{i} + (\dot{r}\sin\theta + \dot{\theta}r\cos\theta)\hat{j} + \dot{z}\hat{k}$

$v_{B/O} = \dot{r}\hat{r} + \dot{z}\hat{k} + \Omega r\hat{\theta}$

$a_{B/O} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \ddot{z}\hat{k} + \ddot{\theta}r\hat{\theta} + \dot{\theta}\dot{r}\hat{\theta} - \dot{\theta}^2 r\hat{r}$

$= (\ddot{r} - \dot{\theta}^2 r)\hat{r} + (\ddot{\theta}r + 2\dot{\theta}\dot{r})\hat{\theta} + \ddot{z}\hat{k}$

$\therefore \ddot{r} = \ddot{\theta} = \ddot{z} = 0$

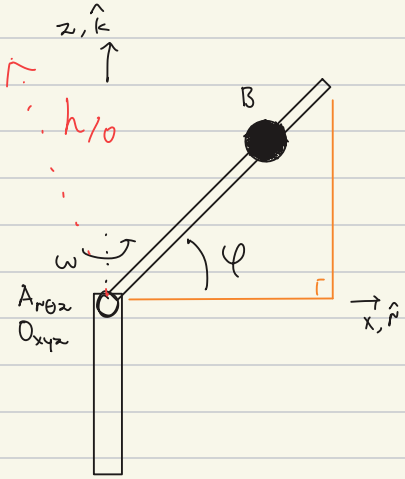
$a_{B/O} = -r\dot{\theta}^2\hat{r} + 2\dot{\theta}\dot{r}\hat{\theta}$

$a_{B/O} = -r\dot{\theta}^2(\cos\theta\hat{i} + \sin\theta\hat{j}) + 2\dot{\theta}\dot{r}(-\sin\theta\hat{i} + \cos\theta\hat{j})$

Made with Goodnotes

$h_{/O} = -\hat{k} - -\hat{r}$

b) to find $h_{/O}$ of monkey.



$h_{/O} = r_{B/O} \times P_{/O}$

$= (r\hat{r} + z\hat{k}) \times m\, v_{B/O}$

$= (r\hat{r} + z\hat{k}) \times m\,(\dot{r}\hat{r} + \dot{z}\hat{k} + r\dot{\theta}\hat{\theta})$

$(r\hat{r} + z\hat{k}) \times (\dot{r}\hat{r} + \dot{z}\hat{k}) = 0$

as they point into the same direction

$h_{/O} = m\,(r\hat{r} + z\hat{k}) \times (r\dot{\theta}\hat{\theta})$

$= mr^2\dot{\theta}\hat{k} - mzr\dot{\theta}\hat{r}$

c) find $\frac{d}{dt} h_{/O}$

$h_{/O} = mr^2\dot{\theta}\hat{k} - mzr\dot{\theta}\hat{r}$

$\frac{d}{dt} h_{/O} = 2mr\dot{r}\dot{\theta}\hat{k} + mr^2\ddot{\theta}\hat{k} - m\dot{z}r\dot{\theta}\hat{r} - mz\dot{r}\dot{\theta}\hat{r} - mzr\ddot{\theta}\hat{r} - mzr\dot{\theta}^2\hat{\theta}$

$= (2mr\dot{r}\dot{\theta} + mr^2\ddot{\theta})\hat{k} - (m(\dot{z}r + z\dot{r})\dot{\theta} - mzr\ddot{\theta})\hat{r} - mzr\dot{\theta}^2\hat{\theta}$

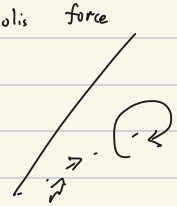
As $\ddot{\theta} = 0$

$\frac{d}{dt} h_{/O} = 2mr\dot{r}\dot{\theta}\hat{k} - m(\dot{z}r\dot{\theta} + z\dot{r}\dot{\theta})\hat{r} - mzr\dot{\theta}^2\hat{\theta}$

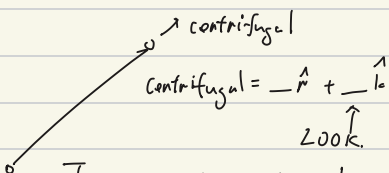
(first) (second) (third)

The first term is the amount of torque neccesary to increase the angular momentum of the body.

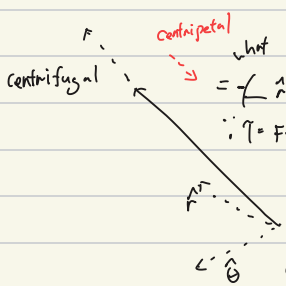
The second term is moment created by coriolis force



Third term is the torque requird to counter the moment generated by centrifugal force of monkey.



To combat the centrifugal force trying to rip the rod upwards.



what we need. We exert a $-\hat{\theta}$ torque ors o

$= -(-\hat{r} + -\hat{k})$ centripetal force

$\therefore T = F \times r_{/C} = -(-\hat{r} + -\hat{k}) \times -\hat{r}$

$= -\hat{\theta}$

So its essentially the centripetal force!

Made with Goodnotes